Progressive Electricity Billing System

Problem Statement: Calculate household electricity bills using multiple consumption slabs, fixed charges, surcharges and rebates. Compare bills across several months and identify unusual increases.

1. Computational Thinking

Computational thinking means solving a problem step by step using logical methods.

1. Decomposition

- Get the number of months.
- Enter electricity units consumed each month.
- Calculate charges based on different slabs.
- Add fixed charges and surcharges.
- Apply rebates.
- Compare monthly bills.

2. Pattern Recognition

- Identify repeated electricity consumption patterns.
- Observe increases or decreases in monthly bills.
- Identify unusual increases.

3. Abstraction

- Consider only important details such as units, slab rates, fixed charges, surcharges and rebates.

4. Algorithm Design

- Develop step-by-step instructions to calculate and compare bills.

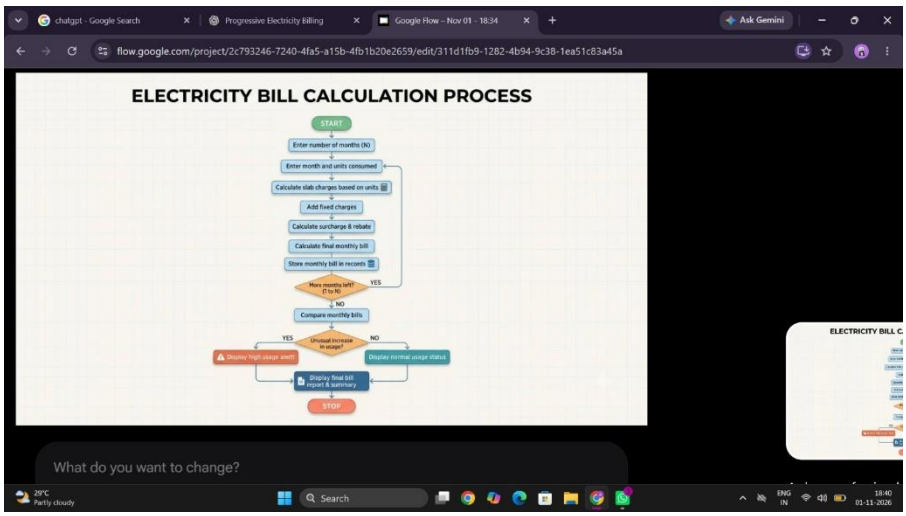
2. Algorithm

1. Start.
2. Read the number of months.
3. Set slab rates and fixed charges.
4. Read electricity units for each month.
5. Calculate energy charges:
    - First 100 units: ₹2 per unit.

- Next 100 units: ₹3 per unit.
- Above 200 units: ₹5 per unit.

6. Add fixed charge of ₹50.
7. Calculate 5% surcharge.
8. Apply a rebate of ₹20 if consumption is below 100 units.
9. Calculate the final bill.
10. Store monthly bills in a list.
11. Compare consecutive monthly bills.
12. If the bill increases by more than 30%, display "Unusual Increase".
13. Display all monthly bills.
14. Stop.

3. Flowchart:



Flowchart symbols:

Symbol	**Purpose**
Oval	Start / Stop
Parallelogram	Input / Output
Rectangle	Calculation
Diamond	Condition
Arrow	Flow direction

4. Variables

Variable	Purpose
months	Number of months
units	Electricity consumed
energy	Energy charges
fixed	Fixed charge
surcharge	Extra charge
rebate	Discount
bill	Final bill
bills	Stores monthly bills
increase	Bill increase percentage
i	Loop counter

5. Data Types

Data type	Variables	Example
Integer	months, units	3, 150
Float	energy, bill	450.50
String	status	"Normal"
List	bills	[350, 450, 600]
Tuple	slab rates	(2, 3, 5)
Boolean	unusual	True / False

6. Operators

Operator	Purpose	Example
+	Addition	energy + fixed
-	Subtraction	bill - rebate
*	Multiplication	units * rate

/	Division	total / 100
>	Greater than	bill > 500
<	Less than	units < 100
==	Equal to	months == 3
and	Logical AND	units > 100 and units < 200

7. Conditional Statements

Use if, elif, and else to calculate slab charges.

if units <= 100: energy = units * 2elif units <= 200: energy = 100 * 2 + (units - 100) * 3else: energy = 100 * 2 + 100 * 3 + (units - 200) * 5

8. Loops

Use a for loop to calculate bills for multiple months.

for i in range(months): units = int(input("Enter units: ")) print("Month:", i + 1)

9. Strings

Strings are used to display messages and bill status.

status = "Normal"print("Electricity Bill:", status)

10. Pattern Generation

Example: Display a monthly bill pattern.

for i in range(1, 4): print("*" * i)

Output:

*

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11. Lists

Lists store electricity bills for different months.

bills = [350, 450, 600]print(bills)print("Highest bill:", max(bills))print("Lowest bill:", min(bills))

12. Tuples

Tuples store fixed electricity slab rates.

rates = (2, 3, 5)print(rates)

13. Packing and Unpacking

Packing: Storing multiple values in a tuple.

bill_details = (150, 550, 50)

Unpacking: Separating tuple values into variables.

units, energy, fixed = bill_detailsprint(units)print(energy)print(fixed)

14. Pseudocode

START

INPUT months

SET rates = (2, 3, 5)

SET bills = empty list

FOR each month from 1 to months

  INPUT units

  IF units <= 100

    energy = units * 2

  ELSE IF units <= 200

    energy = 200 + (units - 100) * 3

  ELSE

    energy = 500 + (units - 200) * 5

  END IF

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    fixed = 50
    surcharge = energy * 0.05

    IF units < 100
        rebate = 20
    ELSE
        rebate = 0
    END IF

    bill = energy + fixed + surcharge - rebate

    ADD bill to bills

END FOR

FOR i from 1 to months - 1

    increase = ((bills[i] - bills[i-1])
                / bills[i-1]) * 100

    IF increase > 30
        PRINT "Unusual Increase"
    ELSE
        PRINT "Normal"
    END IF

END FOR
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PRINT bills

STOP

15. Python Program

months = int(input("Enter number of months: "))bills = []rates = (2, 3, 5)for i in range(months): units = int(input("Enter electricity units: ")) if units <= 100: energy = units * rates[0] elif units <= 200: energy = 100 * rates[0] + (units - 100) * rates[1] else: energy = (100 * rates[0] + 100 * rates[1] + (units - 200) * rates[2]) fixed = 50 surcharge = energy * 0.05 if units < 100: rebate = 20 else: rebate = 0 bill = energy + fixed + surcharge - rebate bills.append(bill) print("Month", i + 1, "Bill:", round(bill, 2))for i in range(1, months):

16. Sample Output

Enter number of months: 3

Enter electricity units: 80

Month 1 Bill: 198.0

Enter electricity units: 120

Month 2 Bill: 365.0

Enter electricity units: 250

Month 3 Bill: 680.0

Month 2 : Unusual Increase

Month 3 : Unusual Increase

All Monthly Bills: [198.0, 365.0, 680.0]

17. Conclusion

The Progressive Electricity Billing System calculates electricity bills using slab rates, fixed charges, surcharges and rebates. It uses computational thinking, algorithms, flowcharts,

variables, data types, operators, conditional statements, loops, strings, patterns, lists, tuples and packing/unpacking to identify unusual increases in monthly bills.